

Q1 (10 points)

A fair coin is tossed three times. Let X be the number of heads obtained.

- (a) Find the PMF of X .
- (b) Plot the PMF.
- (c) Compute $P(X = 2)$ and $P(X \geq 1)$.

for binomial distribution. $n=3$, $p=0.5$

a)

$$P(X=k) = \binom{3}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{3-k} = \frac{1}{8} \binom{3}{k}$$

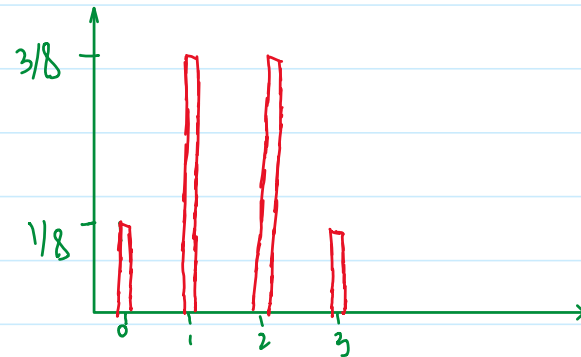
- for each value of X :-

$$X=0: P(X=0) = \frac{1}{8} \binom{3}{0} = \frac{1}{8} \quad b)$$

$$X=1: P(X=1) = \frac{1}{8} \binom{3}{1} = \frac{3}{8}$$

$$X=2: P(X=2) = \frac{1}{8} \binom{3}{2} = \frac{3}{8}$$

$$X=3: P(X=3) = \frac{1}{8} \binom{3}{3} = \frac{1}{8}$$



$$c) P(X=2) \text{ \& } P(X \geq 1)$$

$$P(X \geq 1) = 1 - P(X=0) = 1 - \frac{1}{8} = \frac{7}{8} = 0.875$$

Q2 (10 points)

A discrete random variable X takes values 0, 1, 2, 3 with

$$p_X(k) = P(X = k) = c(2k + 1), \quad k = 0, 1, 2, 3.$$

- (a) Determine c .
- (b) Compute $P(X > 1)$.
- (c) Find the Mean of X .
- (c) Find the Variance of X .

$$a) \sum_{k=0}^3 P(X=k) = 1$$

So,

$$c [(2 \cdot 0 + 1) + (2 \cdot 1 + 1) + (2 \cdot 2 + 1) + (2 \cdot 3 + 1)] = 1$$

$$c [1 + 3 + 5 + 7] = 1$$

$$16c = 1$$

$$c = 1/16$$

$$b) P(X > 1) :-$$

$$P(X > 1) = P(X=2) + P(X=3)$$

$$P(X=2) = c (2 \cdot 2 + 1) = \frac{1}{16} \cdot 5 = \frac{5}{16}$$

$$P(X=3) = \frac{1}{16} (6+1) = \frac{7}{16}$$

So,

$$P(X > 1) = \frac{5}{16} + \frac{7}{16} = \frac{12}{16} = \frac{3}{4}$$

So,

$$P(X > 1) = \frac{5}{16} + \frac{7}{16} = \frac{3}{4}$$

c) Mean of X :-

$$E[X] = \sum_{k=0}^3 k \cdot P(X=k)$$

$$P(X=0) = \frac{1}{16} \quad \text{So,}$$

$$P(X=1) = \frac{3}{16} \quad E[X] = 0 \cdot \frac{1}{16} + 1 \cdot \frac{3}{16} + 2 \cdot \frac{5}{16} + 3 \cdot \frac{7}{16}$$

$$P(X=2) = \frac{5}{16} \quad = 0 + \frac{3}{16} + \frac{10}{16} + \frac{21}{16} = \frac{17}{8}$$

$$P(X=3) = \frac{7}{16}$$

d) Variance of X :-

$$E[X^2] = \sum_{k=0}^3 k^2 \cdot P(X=k)$$

$$E[X^2] = 0^2 \cdot \frac{1}{16} + 1^2 \cdot \frac{3}{16} + 2^2 \cdot \frac{5}{16} + 3^2 \cdot \frac{7}{16} = \frac{43}{8}$$

$$\text{Var}(X) = E[X^2] - E[X]^2$$

$$\text{Var}(X) = \frac{43}{8} - \left(\frac{17}{8}\right)^2 = \frac{43}{8} - \frac{289}{64} = 0.859$$

Q3 (10 points)

A coin is tossed repeatedly until a Head occurs. The probability of obtaining a Head on a single toss is p . Let the discrete random variable X denote the number of tosses required for a Head to appear for the first time.

- (a) What is the PMF of the random variable X .
- (b) Show that $E[X] = \frac{1}{p}$.
- (c) Find $\text{Var}(X)$.

a) PMF of X :-

let $k-1$ = tails

and probability of head = p

so, for tails probability = $1-p$

PMF:

$$P(X=k) = (1-p)^{k-1} \cdot p, \text{ for } k=1, 2, 3, \dots, \infty$$

b) using formula of geometric series:-

$$E[X] = \sum_{k=1}^{\infty} k \cdot p(1-p)^{k-1}$$

$$E[X] = p \sum_{k=1}^{\infty} k \cdot (1-p)^{k-1}$$

let $q = 1-p$.

$$\therefore \frac{1}{1-r}, r < 1$$

$$\sum_{k=1}^{\infty} k q^{k-1} = \frac{d}{dq} \left(\sum_{k=0}^{\infty} q^k \right) = \frac{d}{dq} \left(\frac{1}{1-q} \right) = \frac{1}{(1-q)^2}$$

So,

$$E[X] = \frac{1}{(1-q)^2} \cdot p = p \cdot \frac{1}{(1-(1-p))^2} = \cancel{p} \frac{1}{p^2} = \boxed{1/p} \text{ Ans.}$$

c) Find $\text{Var}(X)$:-

$$E[X^2] = \sum_{k=1}^{\infty} k^2 \cdot p(1-p)^{k-1}$$

$$\sum_{k=1}^{\infty} k^2 q^{k-1} = \frac{1+q}{(1-q)^3}$$

with $q = 1-p$:-

$$E[X^2] = \frac{1 + (1-p)}{(1 - (1-p))^3} = p \cdot \frac{2-p}{p^3} = \frac{2-p}{p^2}$$

Now,

$$\text{Var}(X) = E[X^2] - E[X]^2$$

$$= \frac{2-p}{p^2} - \left(\frac{1}{p}\right)^2 = \frac{2-p-1}{p^2} = \boxed{\frac{1-p}{p^2}} \text{ Ans.}$$

Q4 (10 points)

From the Bernoulli, Binomial, Geometric, Uniform, and Poisson random variables, which one will be the most appropriate choice to model the following:

- Number of games Karachi Kings will win in the 2026 edition of Pakistan Super League
- Outcome of rolling a fair 6-sided die
- Number of balls that will be bowled by a new player in international cricket before getting his/her first wicket.

a. Binomial distribution

- fixed number of games in (PSL) series.
- each match is win or lose (lose).
- in each game the probability of winning is equal.
- games are independent trials.

b. Uniform distribution

- fixed number of equally liked outcomes.

b. Uniform distribution

- fixed number of equally liked outcomes.
- each outcome has same probability i.e. $1/6$
- single trial

c. Geometric distribution:

- Counting number of trials until first success
- each ball has constant probability of hitting wicket.
- we stop at first success

Q5 (10 points)

- a) One out of every eight phone calls to your house is a telemarketer. Assume that the telemarketers' calls are independent. Let X denote the number of telemarketers during the next three calls. What is the PMF of X ?
- b) One out of every eight phone calls to your house is a telemarketer. Assume that the telemarketers' calls are independent. Let X denote the number of phone calls until the telemarketer calls (If the second call is a telemarketer, $X = 2$ and so on). What is the probability that $X \geq 3$?

1) call from telemarketer = $1/8$

• Not from telemarketer = $7/8$

• $n=3$

- let X be the number of telemarketer calls:

with binomial distribution:-

$$P(X=k) = \binom{3}{k} \left(\frac{1}{8}\right)^k \left(\frac{7}{8}\right)^{3-k}, \quad k=0,1,2,3$$

So,

$$P(X=0) = \binom{3}{0} \left(\frac{1}{8}\right)^0 \left(\frac{7}{8}\right)^3 = 1 \cdot 1 \cdot \frac{343}{512} = \frac{343}{512}$$

$$P(X=1) = \binom{3}{1} \left(\frac{1}{8}\right)^1 \left(\frac{7}{8}\right)^2 = 3 \cdot \left(\frac{1}{8}\right) \left(\frac{7}{8}\right)^2 = \frac{3 \cdot 1}{8} \cdot \frac{49}{64} = \frac{147}{512}$$

$$P(X=2) = \binom{3}{2} \left(\frac{1}{8}\right)^2 \left(\frac{7}{8}\right)^1 = 3 \cdot \frac{1}{64} \cdot \left(\frac{7}{8}\right) = \frac{21}{512}$$

- $P(X=2) = \binom{3}{2} \left(\frac{1}{8}\right)^2 \left(\frac{7}{8}\right)^1 = 3 \cdot \frac{1}{64} \cdot \left(\frac{7}{8}\right) = \frac{21}{512}$
- $P(X=3) = \binom{3}{3} \left(\frac{1}{8}\right)^3 \left(\frac{7}{8}\right)^0 = 1 \cdot \frac{1}{512} \cdot 1 = \frac{1}{512}$

b) Probability that $X \geq 3$:

$$P = 1/8$$

$$X \geq 3$$

$$P(X=k) = (1-p)^{k-1} \cdot p$$

So,

$$P(X \geq 3) = \left(\frac{7}{8}\right) \cdot \left(\frac{7}{8}\right) = \frac{49}{64}$$

first two calls are not telemarketers.

Q6 (10 points)

Let the random variable X be the outcome of rolling a fair 6-sided die.

a) What is $E[X^2 + 1]$?

b) What is $E[2X + 1]$?

$$a) E[X^2 + 1] = E[X^2] + E[1] = E[X^2] + 1$$

Now,

$$E[X^2] = \sum_{k=1}^6 k^2 \cdot P(X=k) = \frac{1}{6} \sum_{k=1}^6 k^2$$

$$= \frac{1}{6} (1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) = \frac{91}{6}$$

Now:-

$$E[X^2+1] = \frac{91}{6} + 1 = \boxed{\frac{97}{6} \text{ Ans.}}$$

$$b) E[2X+1] = 2E[X] + 1 =$$

Now,

$$E[X] = \frac{1}{6} (1+2+3+4+5+6) = \frac{7}{2}$$

$$E[2X+1] = \frac{14}{2} + 1 = \boxed{8} \text{ Ans.}$$

Q7 (10 points)

Consider the following random variables X & Y :

X = Magnitude of the difference between the outcomes of two rolls of a fair 3-sided die.

- (If the first roll results in 1 and second roll results in 3, the value taken by X is 2)
- (If the first roll results in 3 and second roll results in 1, the value taken by X is 2)

$Y = 2X$

- Calculate and plot the PMF of X .
- Calculate and plot the PMF of Y .
- Calculate Mean of Y .
- Calculate Variance of Y .

first roll	second roll	Difference
1	1	0
1	2	1
1	3	2
2	1	1
2	2	0
2	3	1
3	1	2
3	2	1

$>$	1	$<$
3	2	1
3	3	0

Now:-

$X=0$, 3 outcomes

$X=1$, 4 outcomes

$X=2$, 2 outcomes.

Total = 9 outcomes.

for,

$$P(X=0) = \frac{3}{9} = \frac{1}{3}$$

$$P(X=1) = \frac{4}{9}$$

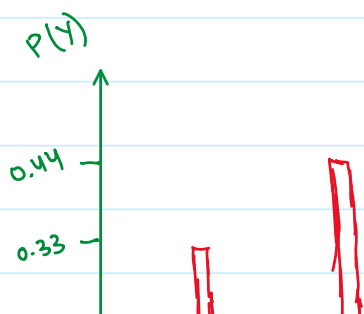
$$P(X=2) = \frac{2}{9}$$

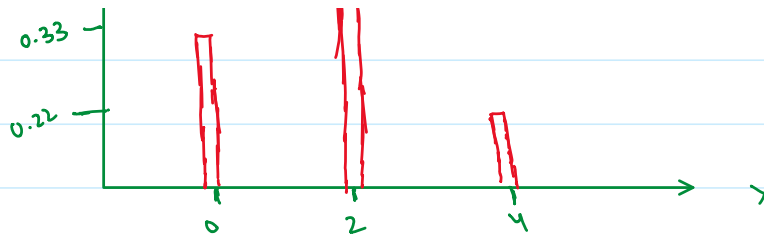
b. $Y = 2X$

$$P(Y=0) = P(X=0) = \frac{1}{3}$$

$$P(Y=2) = P(X=1) = \frac{4}{9}$$

$$P(Y=4) = P(X=2) = \frac{2}{9}$$





c. Mean of y :-

$$E[Y] = \sum_y y \cdot P(Y=y)$$

$$E[X] = 0 \cdot \left(\frac{1}{3}\right) + 2 \cdot \left(\frac{4}{9}\right) + 4 \cdot \left(\frac{2}{9}\right) = \frac{16}{9}$$

d. $\text{Var}(Y) = ?$:-

$$E[Y^2] = 0 \cdot \left(\frac{1}{3}\right) + 4 \cdot \left(\frac{4}{9}\right) + 16 \cdot \left(\frac{2}{9}\right) = \frac{16}{3}$$

$$\text{Var}(Y) = E[Y^2] - E[Y]^2 = \frac{16}{3} - \left(\frac{16}{9}\right)^2 = \frac{176}{81}$$

Q8 (10 points)

Suppose you roll a fair six-sided Red die and a fair tetrahedral Green die. Consider the following two random variables X & Y :

X = number shown on the fair six-sided Red die

$$Y = \begin{cases} 0 & \text{if the two dice show the same number,} \\ 1 & \text{if the number on the green tetrahedral die is bigger than the number on the red die,} \\ 2 & \text{if the number on the red six-sided die is bigger than the number on the green die.} \end{cases}$$

- What is $p_{X,Y}(x,y)$? (You can write it in table form).
- What is $p_Y(y)$?
- What is $E[X]$?
- What is $p_{X|Y}(x|2)$?

a.	Red (X)	Green	Y value
	1	1	0
		2	1
		3	1
		4	1
	2	1	2
		2	0
		3	1
		4	1
	3	1	2
		2	2
		3	0
		4	1
	4	1	2
		2	2
		3	2
		4	0
	5	1	2
		2	2
		3	2
		4	2
	6	1	2
		2	2
		3	2
		4	2

• Joint PMF:

X	X=0	X=1	X=2
X=1	1/24	3/24	0
X=2	1/24	2/24	1/24
X=3	1/24	1/24	2/24

$X=2$	$1/24$	$2/24$	$1/24$
$X=3$	$1/24$	$1/24$	$2/24$
$X=4$	$1/24$	0	$3/24$
$X=5$	0	0	$4/24$
$X=6$	0	0	$4/24$

b) $P_Y(Y) = ?$

$$P(Y=0) = \frac{1+1+1+1}{24} = \frac{4}{24} = \frac{1}{6}$$

$$P(X=1) = \frac{3+2+1+0}{24} = \frac{1}{4}$$

$$P(Y=2) = \frac{0+1+2+3+4+4}{24} = \frac{7}{12}$$

c. $E[X] = \sum_{x=1}^6 x \cdot P(X=x) = \frac{1}{6} (1+2+3+4+5+6) = \frac{7}{2}$

d. $P_{X|Y}(x|2):$

so,

$$P_{X|Y}(1|2) = \frac{0}{7/12} = 0$$

$$P_{X|Y}(2|2) = \frac{1/24}{7/12} = \frac{1/24 \cdot 12}{7} = \frac{1}{14}$$

$$P_{X|Y}(3|2) = \frac{2/24}{7/12} = \frac{2}{14} = \frac{1}{7}$$

$$P_{x_1, x_2}(4|2) = \frac{3/24}{7/12} = \frac{3}{14}$$

$$P_{x_1, x_2}(5|2) = \frac{4/24}{7/12} = \frac{4}{14} = \frac{2}{7}$$

$$P_{x_1, x_2}(6|2) = \frac{4/24}{7/12} = \frac{4}{14} = \frac{2}{7}$$

Q9 (10 points)

A Random Variable Z_n is defined as $Z_n = \min(X_1, X_2, \dots, X_n)$, where X_i 's are outcomes of independent rolls of fair 3-sided die. As an example, $Z_2 = \min(X_1, X_2)$ - meaning Z_2 is the minimum of the outcome of two independent rolls of fair 3-sided die X_1 and X_2 .

a) What is the mean of Z_2 ?

b) Let's define a random variable Y as $Y = 2Z_1 + Z_2$. What is the mean of Y ?

a)

X_1	X_2	$\min(X_1, X_2)$
1	1	1
1	2	1
1	3	1
2	1	1
2	2	2
2	3	2
3	1	1
3	2	2
3	3	3

PMF of Z_2 :-

$$P(Z_2=1) = 5/9$$

$$P(Z_2=2) = 3/9 = 1/3$$

$$P(Z_2=3) = 1/9$$

Mean of Z_2 :-

$$E[Z_2] = 1 \cdot \frac{5}{9} + 2 \cdot \frac{3}{9} + 3 \cdot \frac{1}{9} = \frac{14}{9}$$

b. Find the mean of Y , where $Y = 2Z_1 + Z_2$

$$E[Z_1] = m[X_1]$$

$$E[Z_1] = \frac{1+2+3}{3} = \frac{6}{3} = 2$$

$$E[Y] = 2(2) + \frac{14}{9} = \boxed{\frac{50}{9}} \text{ Ans}$$

Q10 (10 points)

Consider four independent rolls of a fair 6-sided die. Let X be the number of 1's and let Y be the number of 2's obtained. What is the joint PMF of X and Y ? (You can give your answer in tabular form).

X = 1's obtained Y = 2's obtained

- each try can show either '1' or '2' or 'other'.
- $P(\text{roll}=1) = 1/6$
- $P(\text{roll}=2) = 1/6$

- $P(\text{roll} = \text{other}) = 4/6$

- $X + Y \leq 4$

$$P(X=x, Y=y) = \frac{4!}{x! \cdot y! \cdot (4-x-y)!} \cdot \left(\frac{1}{6}\right)^x \left(\frac{1}{6}\right)^y \left(\frac{4}{6}\right)^{4-x-y}$$

Total trials
failed to get x or y

success of x
success of y

Joint PMF table -

$P_{(x,y)}(x,y)$	$Y=0$	$Y=1$	$Y=2$	$Y=3$	$Y=4$
$X=0$	$256/1296$	$256/1296$	$96/1296$	$16/1296$	0
$X=1$	$256/1296$	$192/1296$	$48/1296$	$4/1296$	0
$X=2$	$96/1296$	$48/1296$	$6/1296$	0	0
$X=3$	$16/1296$	$4/1296$	0	0	0
$X=4$	$1/1296$	0	0	0	0

Q11 (10 points)

Let X be the number of heads obtained when three fair coins are tossed. Given that at least one head occurs, find:

- The conditional PMF of X given that at least one head occurs, i.e. $p_{X|A}(x)$, where $A = \{X \geq 1\}$.
- The conditional expectation $E[X | A]$.

PMF of X :-

$$P(X=0) = C(3,0) \left(\frac{1}{2}\right)^3 = 1/8$$

$$P(X=1) = C(3,1) \left(\frac{1}{2}\right)^3 = 3/8$$

$$P(X=2) = C(3,2) \left(\frac{1}{2}\right)^3 = 3/8$$

$$P(X=2) = C(3,2) (1/2)^3 = 3/8$$

$$P(X=3) = C(3,3) (1/2)^3 = 1/8$$

$$a) P(A) = P(X \geq 1) = 1 - P(X=0) = 1 - 1/8 = 7/8$$

for conditions.

$$P_{X|A}(x) = \frac{P(X=x, A)}{P(A)}$$

As $X=0$ doesn't satisfy 'A',

So,

$$P_{X|A}(0) = 0$$

$$P_{X|A}(1) = \frac{3/8}{7/8} = \frac{3}{7}$$

$$P_{X|A}(2) = \frac{3/8}{7/8} = \frac{3}{7}$$

$$P_{X|A}(3) = \frac{1/8}{7/8} = \frac{1}{7}$$

$$b) E[X|A] = \sum_{x=0}^3 x \cdot P_{X|A}(x)$$

$$= 0 \cdot 0 + 1 \cdot \frac{3}{7} + 2 \cdot \frac{3}{7} + 3 \cdot \frac{1}{7} = \boxed{\frac{12}{7}} \text{ Ans.}$$

Q12 (10 points)

Let's assume that 40% of current HU students belong to AHSS and 60% belong to DSSE. Let X be the random variable that a randomly selected HU student belongs to DSSE. (X takes on value 1 if the student belongs to DSSE and 0 otherwise.) Let Y be the random variable that captures the CGPA of a randomly selected HU student. You have been provided the following information:

$$E[Y|X=1] = 3.0 \quad E[Y|X=0] = 3.2$$

Consider the following exercise: Stand at the main entrance gate of HU, ask the CGPA of first 100 students, and calculate their average CGPA. Before conducting the above-mentioned exercise, what will be your best guess for the average CGPA obtained through the exercise? Justify your answer by pointing out the relevant laws/theorems discussed in this course?

(Note: No points will be awarded for an answer without a valid justification and identification of the appropriate theorem/laws discussed in the course).

$$0.4 = \text{Ahss} \rightarrow X=0$$

$$0.6 = \text{Dsse} \rightarrow X=1$$

$$E[Y|X=1] = 3.0$$

$$E[Y|X=0] = 3.2$$

expectation =

$$\begin{aligned} E[Y] &= E[Y|X=1] P(X=1) + E[Y|X=0] P(X=0) \\ &= 3.0 \times 0.6 + 3.2 \times 0.4 = 3.08 \end{aligned}$$

Expected value of mean is 3.08

- After asking from 100 students:- (By linearity expectation)

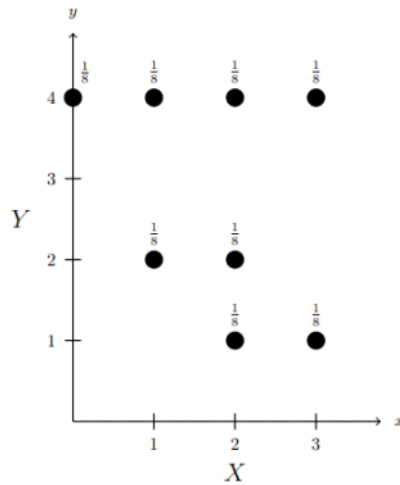
$$E[\bar{Y}] = E\left[\frac{1}{100} \sum_{i=1}^{100} Y_i\right] = \frac{1}{100} \times 100 \times E[Y] = E[Y]$$

Therefore:- $E[\bar{Y}] = 3.08$

- Best guess will be that you will be 3.08, by the law of Total expectation, which gives us the overall expected CGPA of Any randomly selected HU student, and the linearity of expectation which tells us that sample mean is equals to population mean.

Q13 (10 points)

Consider the following joint PMF of two random variables X and Y :



- Are X and Y independent?
- Draw the following: $p_{X|Y}(x|1)$, $p_{Y|X}(y|1)$.
- What is $E[X|Y=4]$?
- Let $R = \max(X, Y)$, what is the mean of R ?

a) For $p_X(x)$

$$p_X(0) = P(0, 4) = \frac{1}{8}$$

$$p_X(1) = P(1, 4) + P(1, 2) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

$$p_X(2) = P(2, 4) + P(2, 2) + P(2, 1) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

For $p_Y(y)$

$$p_Y(4) = 4\left(\frac{1}{8}\right) = \frac{1}{2}$$

$$p_Y(2) = 2\left(\frac{1}{8}\right) = \frac{1}{4}$$

$$p_Y(1) = 2\left(\frac{1}{8}\right) = \frac{1}{4}$$

- For independence

- For independence

$$P_{x,y}(0,4) = P_x(0) \cdot P_y(4)$$

$$\frac{1}{8} = \frac{1}{8} \cdot \frac{1}{2}$$

$$\frac{1}{8} \neq \frac{1}{16}$$

So,

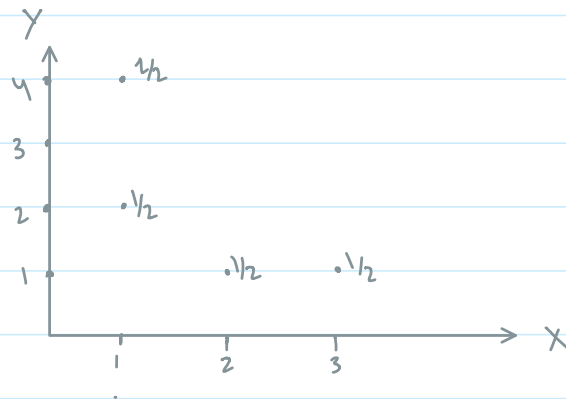
X & Y are not independent.

$$b) P_{x|y}(2|1) = \frac{1/8}{1/4} = \frac{1}{2}$$

$$P_{x|y}(3|1) = 1/2$$

$$P_{y|x}(4|1) = \frac{1/8}{1/4} = \frac{1}{2}$$

$$P_{y|x}(2,1) = 1/2$$



$$c) E[X|Y=4] = \sum_{x=0}^3 x \cdot \frac{1}{4}$$
$$= \frac{1}{4} (0+1+2+3) = \frac{3}{2}$$

$$d. R = \max(X, Y)$$

$$(0, 4) \Rightarrow r=4$$

$$(1, 4) \Rightarrow r=4$$

$$(2, 4) \Rightarrow r=4$$

$$(3, 4) \Rightarrow r=4$$

$$(1, 2) \Rightarrow r=2$$

$$(2, 2) \Rightarrow r=2$$

$$(2, 1) \Rightarrow r=2$$

$$(3, 1) \Rightarrow r=3$$

$$(2,1) \Rightarrow Y=2$$

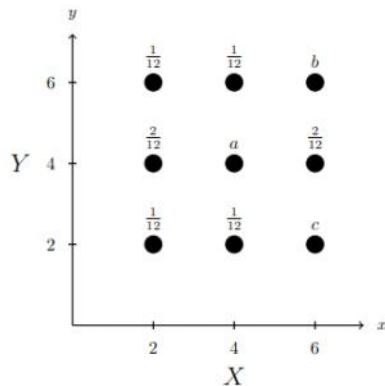
$$(3,1) \Rightarrow Y=3$$

• For $E[R]$,

$$E[R] = \frac{1}{8} (4 + 4 + 4 + 4 + 2 + 2 + 2 + 3) = \frac{25}{8}$$

Q14 (10 points)

Let X and Y be two discrete random variables with their joint PMF shown below:



- Calculate $p_X(2)$.
- Calculate $E[Y|X=2]$.
- Determine the values of a , b , and c , under which X and Y are independent random variables. Justify your answer.

$$a) p_X(2) = \frac{1}{12} (1 + 2 + 1) = \frac{1}{3}$$

$$b) E[Y|X=2] :-$$

$$P(Y=2|X=2) = \frac{1/12}{4/12} = \frac{1}{4}$$

$$P(Y=4|X=2) = \frac{2/12}{4/12} = \frac{1}{2}$$

$$P(Y=6|X=2) = \frac{1/12}{4/12} = \frac{1}{4}$$

$$c) a=? \quad b=? \quad c=?$$

Rows:-

c) $u=1, v=1, w=1$
 Rows:-

$$R_2 = \frac{2+c}{12} \quad R_4 = \frac{4+a}{12} \quad R_6 = \frac{2+b}{12}$$

for columns:-

$$C_2 = \frac{4}{12} \quad C_4 = \frac{2+a}{12} \quad C_6 = \frac{2+b+c}{12}$$

- as we know that Independence require both of the numeration n_{ij} to equal to 1:-

So,

$$1 = \frac{C_2 R_2}{12}$$

$$1 = \frac{4(2+c)}{12}$$

so,

$$\boxed{c=1}$$

Now,

$$1 = \frac{C_4 R_4}{12}$$

$$1 = \frac{(2+a)(2+c)}{12}$$

$$\frac{12}{3} = (2+a)$$

$$4-2=a$$

$$\boxed{a=2}$$

c) $P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$ for all x, y

- Marginal PMF of x :-

$$P(X=2) = \frac{1}{12} + \frac{1}{12} + \frac{1}{12} = \frac{4}{12} = \frac{1}{3}$$

$$P(X=4) = \frac{1}{12} + a + \frac{1}{12} = \frac{1}{6} + a$$

$$P(X=6) = b + \frac{2}{12} + c = b + c + \frac{1}{6}$$

- Marginal PMF of y :-

$$P(Y=6) = \frac{1}{12} + \frac{1}{12} + b = \frac{1}{6} + b$$

$$P(Y=4) = \frac{2}{12} + \frac{2}{12} + a = \frac{1}{3} + a$$

$$P(Y=2) = \frac{1}{12} + \frac{1}{12} + c = \frac{2}{12} + c$$

• Probability should equal 1:-

$$\sum_{x,y} P(X=x, Y=y) = 1$$

$$\frac{8}{12} + a + b + c = 1$$

$$\boxed{a + b + c = \frac{1}{3}}$$

• as X and Y are independent:-
($X=4, Y=6$)

$$P(X=4, Y=6) = P(X=4) \cdot P(Y=6)$$

$$\frac{1}{12} = \left(\frac{1}{6} + a\right) \cdot \left(\frac{1}{6} + b\right)$$

for independence ($X=6, Y=6$)

$$P(X=6, Y=6) = P(X=6) \cdot P(Y=6)$$

$$b = \left(b + \frac{1}{6} + c\right) \left(\frac{1}{6} + b\right)$$

• for independence ($X=4, Y=4$)

$$P(X=4, Y=4) = P(X=4) \cdot P(Y=4)$$

$$a = \left(\frac{1}{6} + a\right) \left(\frac{1}{3} + a\right)$$

Solve:-

$$a = \frac{1}{18} + \frac{1}{6}a + \frac{1}{3}a + a^2$$

$$a - \frac{1}{2}a = \frac{1}{18} + a^2$$

$$\frac{1}{2}a = \frac{1}{18} + a^2$$

$$a^2 - \frac{1}{2}a + \frac{1}{18} = 0$$

$$18a^2 - 9a + 1 = 0$$

$$a = \frac{1}{6} \text{ or } \frac{1}{3}$$

- if $a = \frac{1}{3}$, from $a+b+c = \frac{1}{3}$ we get $b+c=0$
so $b=0, c=0$
- $P(X=6) = 0 + \frac{1}{6} + 0 = \frac{1}{6}$ and $P(Y=6) = \frac{1}{6} + 0 = \frac{1}{6}$
- $P(X=6, Y=6) = 0 \neq \frac{1}{6}$ so it doesn't work.
- If $a = \frac{1}{6}$, from $a+b+c = \frac{1}{3}$, we get
 $b+c = \frac{1}{6}$

So in first equation.

$$\frac{1}{12} = \left(\frac{1}{6} + \frac{1}{6}\right) \left(\frac{1}{6} + b\right) = \frac{1}{3} \left(\frac{1}{6} + b\right)$$

So,

$$\frac{1}{12} = \frac{1}{18} + \frac{1}{3}b$$

$$b = \frac{1}{12}$$

Then $c = \frac{1}{6} - \frac{1}{12} = \frac{1}{12}$

$$c = \frac{1}{12}$$

$$a = \frac{1}{6}$$